When Does a Scale Factor Get Squared or Cubed?

Find and fix the scaling, then say which power of the scale factor belongs where

- If every length of a figure is multiplied by k, then every area is multiplied by k^2 and every volume by k^3.
- Round only where a problem tells you to, and always at the last step.

1. A rectangle of area 24 cm^2 is enlarged so that every length is multiplied by a scale factor of 3. Dev writes "area $= 24 \times 3 = 72$ cm^2".

(a) Find the correct area of the enlarged rectangle.

Answer: ________________ cm^2

(b) Explain what Dev did wrong, and say which power of the scale factor belongs in the area calculation.

2. A solid cube has edge 5 cm. Every length of it is multiplied by a scale factor of 4. Find the volume of the enlarged cube.

Answer: ________________ cm^3

3. Two similar cones have heights 6 cm and 15 cm. The smaller cone has volume 90 cm^3. Mia writes "$V = 90 \times \frac{15}{6} = 225$ cm^3".

(a) Find the correct volume of the larger cone.

Answer: ________________ cm^3

(b) Explain what Mia did wrong.

4. Two similar prisms have surface areas 48 cm^2 and 300 cm^2. Find the linear scale factor from the smaller prism to the larger one.

Answer: ________________

5. Two similar solids have volumes 54 cm^3 and 250 cm^3.

(a) Find the linear scale factor from the smaller solid to the larger one, to the nearest hundredth. ________

(b) The smaller solid has surface area 63 cm^2. Find the surface area of the larger solid.

Answer: ________________ cm^2

6. A scale model of a water tank is built at a scale of 1 : 20, so every real length is 20 times the model's.

(a) The model holds 1.5 litres. Find the capacity of the real tank.

Answer: ________________ L

(b) The model's outer shell used 0.8 square metres of sheet metal. Find the sheet metal needed for the real tank.

Answer: ________________ m^2

(c) Explain why the capacity and the sheet-metal area are not multiplied by the same number, even though both come from the single scale factor 20.